UNIVERSITY OF MODENA AND REGGIO EMILIA

"Enzo Ferrari" Department of Engineering

Master's Degree in Artificial Intelligence Engineering (LM-32)

Applying Explainability Methods to Financial Models

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1. Introduction

1.1 Section

This template can be used as a basis for writing the final thesis for the Master's Degree in Computer Engineering/Artificial Intelligence Engineering. The table of contents, as well as the lists of figures and code listings, are generated automatically. The following chapter provides examples of basic elements.

1.1.1 Subsection

This is an example of a subsection.

2. Basic elements

This chapter will contain basic examples of bullet lists, figures, tables, formulas, algorithms, and code.

Example of a bullet list:

- **first bullet point in bold**;
- second bullet point, if you prefer it not to be in bold;

2.1 Section in the Second Chapter

Example of an image in Figure 2.1. Remember to always reference in the text, using \ref, all the figures/tables/algorithms/code you include.[1]

[1] An example of how to add a footnote, often used to cite or include a link to a website, such as `https://ditto.ing.unimore.it/`,

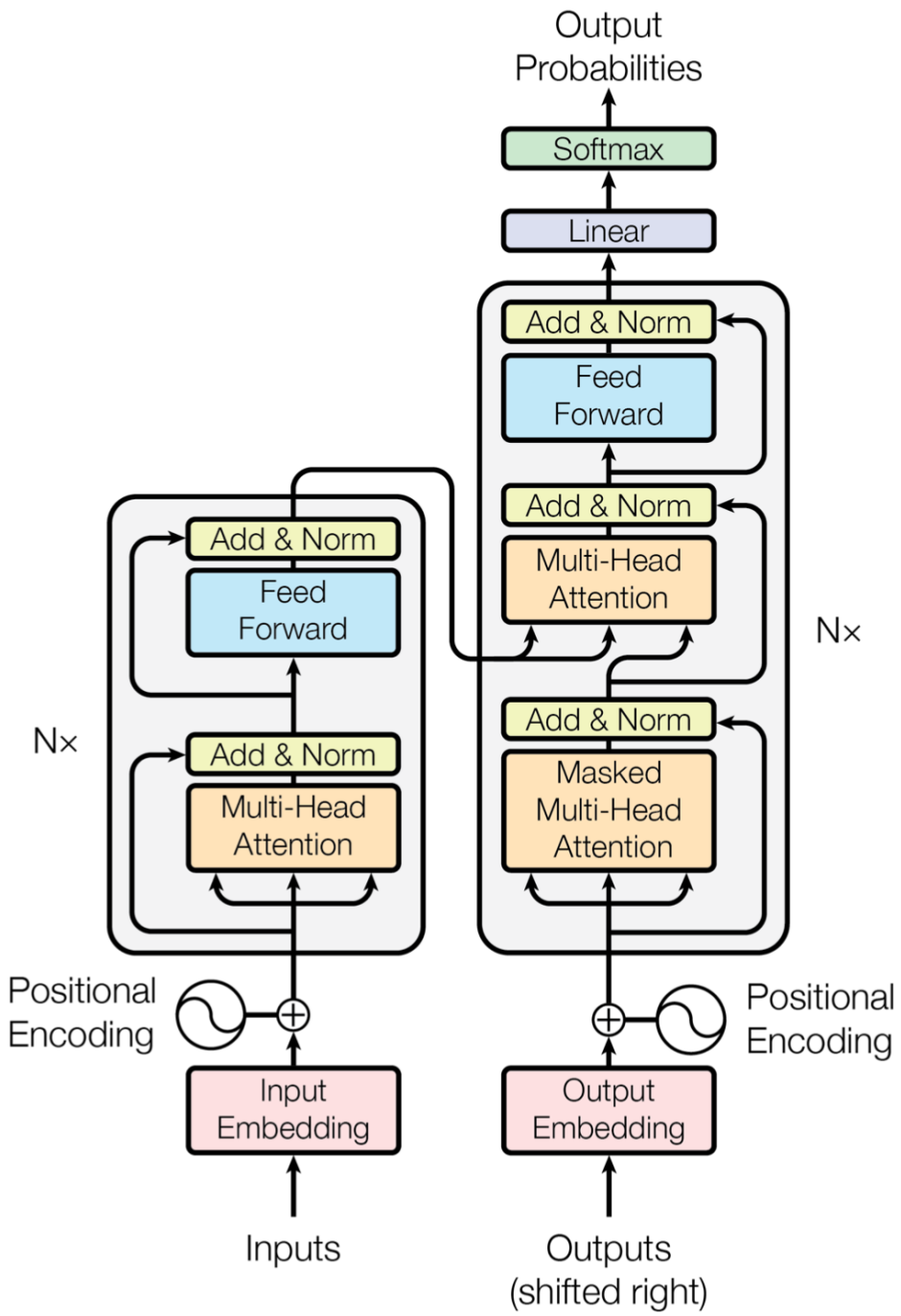

Figure 2.1: The architecture introduced by Vaswani in the famous paper *"Attention is All You Need"* [7].

Table 2.1: Example of a table with various fields, checkmarks, and crosses.

Fruit	**Orange-colored**	**Peelable**	**Weight**	**Price**	**Sugar**	**Rating (1–10)**
Orange	✓	✓	0.200	2.50	9.0	8
Mandarin	✓	✓	0.120	3.20	8.5	9
Lemon	✗	✓	0.100	3.50	2.5	6
Apple	✗	✗	0.180	2.80	10.0	8
Banana	✗	✓	0.150	1.10	12.2	9
Kiwi	✗	✓	0.120	4.00	9.1	7
Mango	✓	✓	0.300	5.00	14.0	10

2.2 Another Section in the Second Chapter

Example of a table with its citation, Table 2.1.

2.3 Mathematical Formulas

Here are two examples of how to write mathematical formulas. As you can easily recognize in Eq. 2.1, we have the integral form of the Fourier Transform, while in Eq. 2.2 we see its discretized version using a summation.

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} \, dt \tag{2.1}$$

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-i \frac{2\pi}{N} kn} \tag{2.2}$$

2.4 Algorithm Example

In Algorithm 1, one way to compute the shortest path can be seen.

Algorithm 1 Dijkstra's Algorithm

```
procedure DIJKSTRA(G, s)
    dist[] ← ∞                                  ▷ Initialize distances to infinity
    dist[s] ← 0                                 ▷ Distance to the source is 0
    Q ← V[G]                                    ▷ Add all nodes to the queue
    while Q is not empty do
        u ← vertex in Q with minimum dist[u]
        remove u from Q
        for each neighbor v of u do
            alt ← dist[u] + length(u, v)
            if alt < dist[v] then
                dist[v] ← alt
                prev[v] ← u
            end if
        end for
    end while
    return dist[], prev[]
end procedure
```

2.5 Code Examples

You may also include some code samples, as long as they are not too long. An example can be seen in Code 2.1.

Code 2.1: Example code, taken from helloworldspam.py

```
print('Starting the ultimate hello world spammer')
for i in range(10000000):
    print("Hello, world!")
```

2.6 Bibliography Management

All bibliographic references must be inserted into the project's ".bib" file. In this template, the file is "bibliography.bib". To cite a reference, use the command "\cite{<id>}", such as [7].

3. Tables and Images

This chapter contains examples of large tables and grid-arranged images.

3.1 1x3 Grid

Here is an example of an image grid, as shown in Figure 3.1.

(a) Orange (b) Apple (c) Banana

Figure 3.1: Photos of fresh fruit in a 1x3 grid.

3.2 2x2 Grid

Here is an example of an image grid, as shown in Figure 3.2.

(a) Orange

(b) Apple

(c) Banana

(d) Pear

Figure 3.2: Photos of fresh fruit in a 2×2 grid.

3.3 Example of a Complex Table

ChapterConclusions

Your conclusions would go here.

Table 3.1: Comparative analysis of different fruits combining various ripening methods, origins, and viewpoints. Run IDs represent different experiments on harvests.

Score	Origin	Fruit	Harvests										
			R1	**R2**	**R3**	**R4**	**R5**	**R6**	**R7**	**R8A**	**R8S**	**R9**	**R10**
Sweetness	All	Apple	87.23	88.01	87.44	86.99	88.12	87.55	88.23	87.66	88.42	88.09	87.77
		Banana	91.31	92.05	91.74	91.92	90.84	92.18	91.95	92.02	91.85	91.99	91.88
	Tropical	Pineapple	84.45	85.02	84.60	84.99	84.25	85.15	84.79	84.92	85.11	84.85	84.88
		Mango	90.12	90.89	91.10	90.67	90.33	91.21	90.97	91.03	91.30	90.95	91.12
	Mediterr.	Orange	88.44	89.11	88.99	89.23	88.51	89.34	89.28	89.21	89.43	89.15	89.32
		Lemon	72.62	73.40	73.88	72.95	72.50	73.91	73.65	73.75	73.84	73.70	73.78
Juiciness	All	Apple	65.42	67.19	66.85	67.00	65.99	67.45	67.11	67.38	67.80	67.21	67.43
		Banana	78.84	79.11	79.67	79.40	78.72	79.91	79.63	79.85	80.10	79.74	79.87
		Grape	82.17	82.69	82.33	82.50	82.02	82.91	82.65	82.84	83.05	82.73	82.91
	Tropical	Mango	85.15	85.44	85.98	85.67	85.01	86.11	85.79	85.92	86.22	85.83	85.96
		Papaya	74.83	75.02	75.37	75.14	74.76	75.49	75.23	75.40	75.60	75.28	75.39
		Pineapple	77.04	77.49	77.71	77.31	77.05	77.80	77.58	77.66	77.93	77.60	77.74
	Mediterr.	Orange	81.90	82.30	82.65	82.49	81.87	82.76	82.52	82.68	82.95	82.59	82.70
		Lemon	61.35	61.88	62.09	61.95	61.24	62.22	62.05	62.17	62.44	62.10	62.26
		Grapefruit	68.75	69.21	69.58	69.37	68.80	69.83	69.65	69.70	69.99	69.68	69.82

Acknowledgements

Here you may include your acknowledgements (ringraziamenti), if you wish.

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